

Hermite interpolation.

Goal: Construct a polynomial that matches not just the points of a function, but its slope (tangents) as well.

Hermite Polynomials.

Definition: The **Hermite polynomial** approximating f is the polynomial $H(x)$ of least degree such that

$$H(x_i) = f(x_i)$$

$$H'(x_i) = f'(x_i)$$

for each $i = 0, 1, \dots, n$.

Remarks: ① The Hermite polynomials $H(x)$ agree with $f(x)$ and the derivatives of the Hermite polynomials $H'(x)$ agree with $f'(x)$.
② The degree of the Hermite Polynomial is $2n+1$ since $2n+2$ conditions must be met.

We will use the **Newton Divided Difference table**, for generating Hermite approximations. This method uses the connection between the n th divided difference and the n th derivative of f

which is evident from the following theorem.

Theorem: Suppose that $f \in C^n[a, b]$ and $x_0, x_1, \dots, x_n$ are distinct numbers in $[a, b]$. Then a number ξ exists in (a, b) with

$$f[x_0, \dots, x_n] = \frac{f^{(n)}(\xi)}{n!}$$

The Method.

Suppose we are given

$$\{(x_0, f(x_0), f'(x_0)), (x_1, f(x_1), f'(x_1)), \dots, (x_n, f(x_n), f'(x_n))\}$$

Define a new sequence $z_0, z_1, z_2, \dots, z_{2n+1}$ by

$$z_{2i} = x_i \quad \text{for } i = 0, 1, \dots, n$$

$$z_{2i+1} = x_i' \quad \text{for } i = 0, 1, \dots, n.$$

Create the divided difference table using $z_0, z_1, \dots, z_{2n+1}$.

Divided Difference Table

z	$f[z]$	First divided differences	Second divided differences
$z_0 = x_0$	$f[z_0] = f(x_0)$	$f[z_0, z_1] = f'(x_0)$	$f[z_0, z_1, z_2] = \frac{f[z_1, z_2] - f[z_0, z_1]}{z_2 - z_0}$
$z_1 = x_0$	$f[z_1] = f(x_0)$	$f[z_1, z_2] = \frac{f[z_2] - f[z_1]}{z_2 - z_1}$	$f[z_1, z_2, z_3] = \frac{f[z_2, z_3] - f[z_1, z_2]}{z_3 - z_1}$
$z_2 = x_1$	$f[z_2] = f(x_1)$	$f[z_2, z_3] = f'(x_1)$	$f[z_2, z_3, z_4] = \frac{f[z_3, z_4] - f[z_2, z_3]}{z_4 - z_2}$
$z_3 = x_1$	$f[z_3] = f(x_1)$	$f[z_3, z_4] = \frac{f[z_4] - f[z_3]}{z_4 - z_3}$	$f[z_3, z_4, z_5] = \frac{f[z_4, z_5] - f[z_3, z_4]}{z_5 - z_3}$
$z_4 = x_2$	$f[z_4] = f(x_2)$	$f[z_4, z_5] = f'(x_2)$	
$z_5 = x_2$	$f[z_5] = f(x_2)$		

$$H_{2n+1}(x) = f[z_0] + \sum_{k=1}^{2n+1} f[z_0, z_1, \dots, z_k] (x - z_0)(x - z_1) \dots (x - z_{k-1})$$

Qn: Find the Hermite Polynomial $H_3(x)$ for the following data:

x	$f(x)$	$f'(x)$
1	2	3
2	6	7

Soln:

$$z_0 = z_1 = 1$$

$$z_2 = z_3 = 2$$

z_0	$f[z_0]$	First	Second	
1	2	3	$\frac{4-3}{2-1} = $ 1	
1	2	$\frac{6-2}{2-1} = 4$		$\frac{3-1}{2-1} = $ 2
2	6		$\frac{7-4}{2-1} = 3$	
2	6	7		

$$H_3(x) = 2 + 3(x-1) + 1(x-1)^2 + 2(x-1)^2(x-2)$$

Qn: Estimate the value of $y(1.05)$ using Hermite interpolation formula from the following data:

x	y	y'
1.00	1.00000	0.5000
1.10	1.04881	0.47673

Soln:

	First	2nd	3rd
$z_0 = 1.00$	1.00000	0.5000	-0.119
$z_1 = 1.00$	1.00000	0.4881	0.0053
$z_2 = 1.10$	1.04881	0.47673	-0.1137
$z_3 = 1.10$	1.04881		

$$H_3(x) = 1 + 0.5(x-1) - 0.119(x-1)^2 + 0.0053(x-1)^2(x-1.10)$$

$$H_3(1.05) = 1.0247$$

Qn:

Use the divided difference approach to approximate $f(0.18)$ given the following data.

x	$f(x)$	$f'(x)$
0.1	-0.29004996	-2.8019975
0.2	-0.56079734	-2.6159201
0.3	-0.81401972	-2.9734038

Soln:

	1st Divided Differences	2nd divided differences	3rd div difference	4th order divided diff	5th order divided diff
$z_0 = 0.1$ $f[z_0] = -0.29004996$	-2.8019975	0.945237	-0.297	-0.47935	-1299.97225
$z_1 = 0.1$ $f[z_1] = -0.29004996$	-2.7074738	0.915537	-0.39287		
$z_2 = 0.2$ $f[z_2] = -0.56079734$	-2.6159201	0.836963	-52.48763	-260.4738	
$z_3 = 0.2$ $f[z_3] = -0.56079734$	-2.5322238	-4.4118			
$z_4 = 0.3$ $f[z_4] = -0.81401972$	-2.9734038				
$z_5 = 0.3$ $f[z_5] = -0.81401972$					

Construct the polynomial:

Using the values from the top diagonal of our table, the 5th degree

Hermite polynomial is :

$$H_5(x) = f[z_0] + f[z_0, z_1](x-z_0) + f[z_0, z_1, z_2](x-z_0)^2 + f[z_0, z_1, z_2, z_3](x-z_0)^2(x-z_2) + f[z_0, z_1, z_2, z_3, z_4](x-z_0)^2(x-z_2)^2 + f[z_0, z_1, z_2, z_3, z_4, z_5](x-z_0)^2(x-z_2)^2(x-z_4)$$

$$\text{i.e., } H_5(x) = -0.29004996 - 2.8019975(x-0.1) + 0.945237(x-0.1)^2 - 0.297(x-0.1)^2(x-0.2) - 0.47935(x-0.1)^2(x-0.2)^2 - 1299.97225(x-0.1)^2(x-0.2)^2(x-0.3)$$

Now find $f(0.18)$ by finding $H_5(0.18)$

$$f(0.18) \approx -0.5077241.$$